

James Condos

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Education

University of Melbourne

Master of Electrical Engineering — Autonomous Systems and AI 2024 – 2026

- Control Algorithms, Embedded Systems, Deep Learning, Optimization, Reinforcement Learning, AI/ML

Bachelor of Science — Major in Mathematical Physics 2019 – 2023

- Advanced Applied Maths, Quantum Physics, Laboratory Physics

Diploma in Computer Science 2019 – 2023

- Java, C, Python Natural Language Processing, AI/ML

Experience

Melbourne Space Program

June 2022 – Present

Lead for ACRUX – 2 Project

Melbourne, VIC

- Lead the Attitude Determination and Control Systems (ADCS) team for ACRUX-2, supporting satellite control-system development ahead of a targeted December 2026 launch.
- Implemented a **C++** finite-state machine to interface reaction wheels and sensors with the on-board computer.
- Contributed to development of **LQR-based** attitude control algorithms for reaction-wheel actuation and spacecraft orientation control.
- Researched and implemented control strategies across sensing and actuation hardware, including B-dot detumbling using a magnetometer and magnetorquers.

Australian Public Service - Science and Technology

May 2025 – Present

STEM Intern Cadetship

VIC, Australia

- A program for high achieving university students to work in Australia's science and technology sector.
- Working on *Python development*, miscellaneous projects and other technical aspects for the department's needs.

PHM Technology

Jan 2025 – July 2025

Electrical Engineering and Research Intern

Melbourne, VIC

- Gained experience with advanced maintenance and reliability software, MADE, which creates digital risk twins.
- Conducted a research project with **Discrete-Event-Simulations (DES)**, **Deep Reinforcement Learning (DRL)** and **Machine Learning (ML)** for multi-agent python simulations. Developed a deterministic decision agent for maintenance operators, which was to be extended to a stochastic framework.

University of Melbourne

Dec 2021 – Jan 2025

Student Researcher for the School of Physics

Melbourne, VIC

- Conducted an independent research project in **quantum machine learning** under academic supervision.
- Investigated links between computation theory and quantum computing, including automata-based models and qubit-state collapse efficiency.
- Explored how quantum-computing concepts could be applied to data-mining and computational-efficiency problems.

Physics Teaching Assistant and Lab Demonstrator

- Supervised undergraduate physics laboratories for classes of 15–20 students.
- Supported student learning during experiments and marked written laboratory reports.

APA Group

Jan 2022 – Mar 2022

Software Developer Intern

Melbourne, VIC

- Built and refined **Java/MySQL** application functionality while supporting CI/CD pipelines with TeamCity and Octopus Deploy to improve release efficiency and system reliability.
- Assisted in delivering software enhancements across development and deployment stages, contributing to more efficient release cycles and improved system reliability.
- Contributed to implementation and deployment activities in a team-based software engineering setting.

Projects and Involvement

Links: [My Portfolio](#) | [Github](#) | [Reference Letter](#)

FPGA-Based Quantum Circuit Simulator (Verilog and C)

- Designed and deployed an **FPGA “engine” that applies quantum gates** to a set of numbers representing a quantum state, focusing on speed and hardware limits.
- Tested accuracy by comparing outputs to a Python/NumPy reference, and recorded latency + resource use to understand trade-offs.
- **Achieved 2–5 μ s per gate at 100 MHz**, and benchmarked end-to-end runtime across small vs larger circuits to show how performance is limited by FPGA memory bandwidth as circuits scale.

Hardware Accelerator for a Convolutional Neural Network (HLS Vitis C++)

- Designed and implemented a hardware accelerator for convolutional neural networks (**CNNs**) using **Vitis HLS and Verilog**, targeting low-latency inference on FPGA hardware.
- Developed and optimised CNN compute kernels in Vitis HLS, applying loop pipelining, unrolling, and memory-interface optimisations to improve throughput and resource efficiency.
- Achieved **$\times 71$ speedup per layer** over software execution by accelerating CNN convolution layers on the FPGA.
- Reduced software execution bottlenecks by offloading computationally intensive CNN layers to custom FPGA hardware.

Telstra’s Advanced Radio Access Network (Final Year Project)

- Leading a final year project optimizing and deploying Telstra’s Advanced Radio Access Network (RAN)
- Employing techniques such as **deep reinforcement learning - still in progress**

Other Notable Projects | *FPGA, Algorithms, Deep Learning, Accelerators*

- Developed a solar cell from scratch, Semiconductor Physics (**Lab work + Solid State Physics**)
- Created an FPGA hardware physical digital clock (**Verilog**)
- Created a Doubly Connected Edge List data structure (**C**)
- Created an autonomous robot for tracking and photographing plants in a room (**ROS2**)

Technical Skills

Languages: Python, C, C++, Java, MATLAB, Verilog, Haskell, JavaScript

Frameworks: React, Node.js, HLS Vitis, CUDA, ROS2

Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, AMD Vitis, Quartus, Atlassian Products